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Seeing Beyond Stellar Interference in the DS Tuc System: Developing a Gaussian Process for Mass Constraint

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ABSTRACT

The process through which planets evolve in the very early stages of their lives has remained somewhat of a black box. Thanks to the *Transiting Exoplanet Survey Satellite* (TESS), however, we are closer to uncovering the mysteries of the early stages of planet formation. The DS Tuc system, located in the Tucana-Horologium young moving group, is only about 40 million years old, and consists of a binary star (DS Tuc A) and a planet between the sizes of Neptune and Saturn (DS Tuc Ab). While DS Tuc's kinematics and some of its physical properties were initially calculated by Newton et al. 2019, the mass of DS Tuc Ab remained unknown. By modeling radial velocity signals from DS Tuc, as well as photometry data collected by both TESS and the Las Cumbres Observatory Network (LCO), we implemented a Gaussian process (GP) to constrain an upper mass limit for DS Tuc Ab: 83 $M_{\oplus}$ (3σ). This illustrates the significance of a GP in using stellar activity indicators to see past the interference from DS Tuc A; for comparison, models that do not account for stellar activity can only yield a much weaker constraint of 220 $M_{\oplus}$.

Keywords: DS Tuc Ab, Exoplanet, TESS, Radial velocity, Photometry, Stellar noise, Mass upper limit.

1. INTRODUCTION

In the DS Tuc system lies a planet slightly larger than Neptune but smaller than Saturn called DS Tuc Ab, discovered by NASA's TESS mission in 2019. This exoplanet—being only about 40 million years old (Newton et al. 2019)—provides a unique opportunity to get a closer look at how planet evolution works in the early stages of a solar system.

Planets the size of DS Tuc Ab are rare and their evolution poorly understood. Theoretical models (e.g. Fortney 2007) and empirical mass-radius relationships (e.g. Chen & Kipping 2016) are poorly constrained by observations at this young age. Because it is so young, DS Tuc Ab is still undergoing drastic evolutionary processes—especially in its atmosphere, from factors such as photoevaporative ultraviolet flux (Newton et al. 2019)—which are causing its measured radius to change. This would naturally affect a mass-radius estimate because as the radius evolves, the mass of DS Tuc Ab remains unchanged. This makes establishing the mass and radius of DS Tuc Ab of extreme importance.

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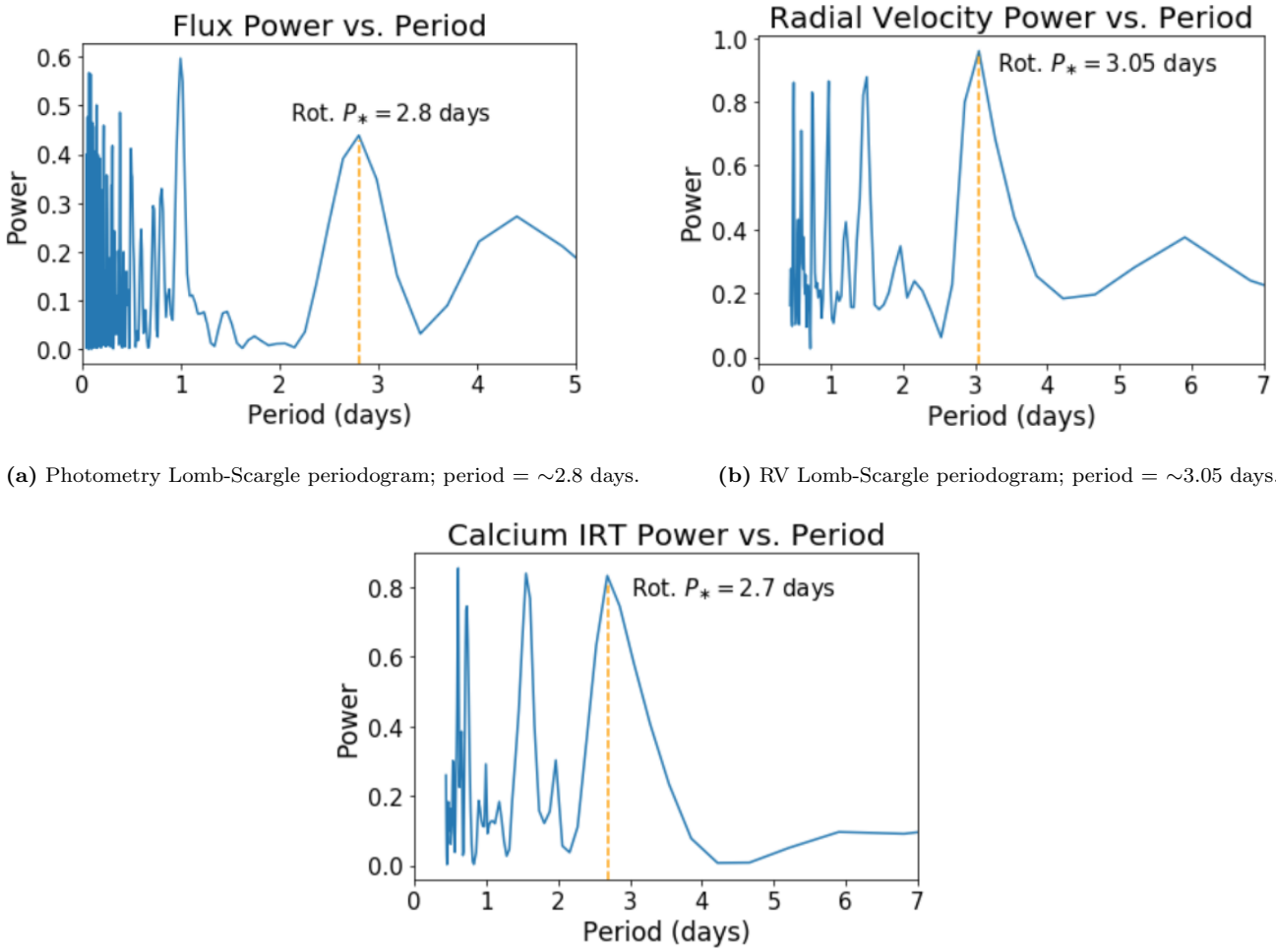
2. BACKGROUND RESEARCH

2.0.1. *Discovery and Early Analysis*

The DS Tuc system was discovered by the *Transiting Exoplanet Survey Satellite* (TESS) during the first science sector of its primary mission, with observations spanning from July to August 2018 (Newton et al. 2019). Its extreme youth provides an opportunity to observe early planetary development before factors such as contraction due to cooling and atmospheric mass loss begin to take effect. However, stellar activity from the host star makes analyzing the planet difficult, with nearly 40% of DS Tuc A being covered in sun spots. This is where taking advantage of multiple discovery



(a) Photometry Lomb-Scargle periodogram; period = ∼2.8 days.

(b) RV Lomb-Scargle periodogram; period = ∼3.05 days.

(c) Calcium IRT Lomb-Scargle periodogram; period = ∼2.7 days.

Figure 1. Photometry, RV, and calcium IRT Lomb-Scargle periodograms demonstrating the datasets that will be fitted together based on how close their periodicities are to the expected value of 2.85 days.

methods—as well as various indicators of stellar activity—proves crucial to obtaining precise data.

Newton et al. 2019 illustrate the process through which DS Tuc Ab was initially discovered and analyzed using a combination of photometric and radial velocity (RV) methods. Gathering data from both TESS and the Las Cumbres Observatory network, they uncovered the planet's radius, $5.17 \pm 0.17\ R_\oplus$, as well as its orbital period, ∼8.1 days, and the star's (DS Tuc A) rotation period of roughly 2.85 days.

3. APPROACH

3.1. *Initial Findings: Lomb-Scargle Analysis*

The first step in developing a model for the DS Tuc data is isolating which activity signatures best describe the observed RV variations. To do this, we take advantage of the Lomb-Scargle periodogram, which is a computationally efficient method of detecting periodic signals in data that is noisy and/or sampled unevenly. This detection is accomplished via least-squares spectral analysis, which is similar to Fourier analysis in the sense that it attempts to fit sinusoids to the given data samples (VanderPlas 2018).

In the case of the DS Tuc data, we search for activity indicators that exhibit periodicity similar to that of the rotation period of DS Tuc A (the star), measured by the discovery TESS observations to be 2.85 days. So we apply `Astropy's` timeseries.LombScargle class to the photometry and RV data from DS Tuc, as well as to $v \cdot \sin i$ and stellar activity indicators calcium IRT and H-alpha. Figure 1 shows the power vs. period plots resulting from the Lomb-Scargle implementation, highlighting the stellar rotation period that each dataset revealed. Calcium IRT resulted in being the best activity indicator, as it produced a rotation period of 2.7 days, very

near that of photometry (flux) and RV, which ended up being 2.8 days and 3.05 days, respectively.

With photometry, RV, and calcium IRT all indicating a rotation period near the desired 2.85 days, we can begin to construct our model for the RV signal from DS Tuc A.

3.2. *Fitting Process*

3.2.1. *Fitting the RV Signals with a Linear Combination of Activity Indicators*

Before going straight into a GP for fitting the datasets, we take a simpler approach by constructing a basic activity model to see if the observed RV signals can be modeled by a linear combination of stellar activity indicators.

The first step in such a fitting process involves a likelihood function, which acts as a measure of the goodness of fit of a model to a given data sample. In this case, our likelihood function will simply be a chi-squared fitness test multiplied by a scalar, as shown in Equation (1).

$$likelihood = -0.5 * \sum \frac{(z - z_{model})^2}{err^2}, \tag{1}$$

where z represents the raw RV data, err is the error in this data, and z_{model} is the RV model we are trying to constrain by isolating unknown values for *hyperparameters* c_0 (for x, the calcium IRT data) and c_1 (for y, the photometry data), and free parameters RV amplitude A (for a sine periodicity term in our model) and added constant K (Equation (2)).

$$z_{model} = c_0 x + c_1 y - A \sin\phi + K, \tag{2}$$

where ϕ represents the phase of the RV dataset, given by:

$$\phi = 2\pi \frac{t_{rv} - t_{transit}}{P_{planet}}, \tag{3}$$

where t_{rv} is the RV timescale, $t_{transit} = 1332.30997 + 2457000\ BJD$ (the transit time of DS Tuc Ab), and P_{planet} is the planet's orbital period of ~8.138268 days (Newton et al. 2019).

To identify the best fitting values of the free parameters, we employ a Markov Chain Monte Carlo (MCMC) analysis, initializing each hyperparameter at a best-guess value and then taking small steps in the direction with the highest probability of reaching the desired value (minimizing error by utilizing our likelihood function). After the hyperparameters are constrained, we simply plug them into our model and plot the model along with the raw RV data, as shown in Figure 2.

Although the MCMC does a good job at constraining the hyperparameters of our model, it struggles at incorporating the calcium IRT data well enough to properly fit the RV data. This shows that the RV stellar activity signal is more complex than can be accounted for by our simple linear model. We therefore seek more complex models to better describe the RV observations.

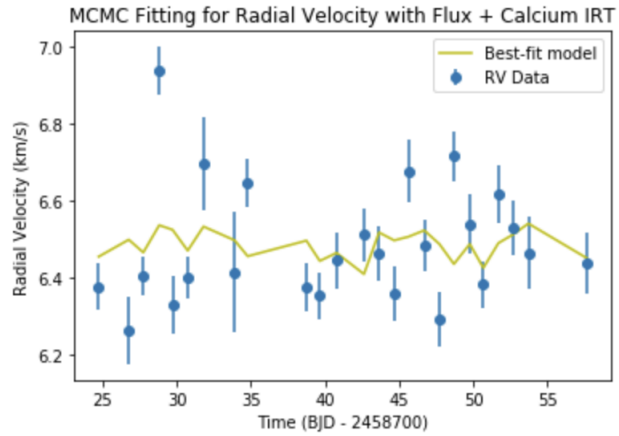


Figure 2. A linear best-fit stellar activity model for the observed RVs. This model does not sufficiently account for the observed stellar variations we see in the RV signal.

3.2.2. *The Gaussian Process*

Fitting conventional models to data from photometry, RVs, calcium II infrared triplets (calcium IRTs), $v \cdot \sin i$, etc. can be difficult in a system with a great deal of stellar noise such as DS Tuc (caused by magnetic activity cycles, granulation phenomena, outer stellar envelope oscillation, etc.). The data mainly affected by stellar activity are the RV measurements, for although RV-measuring instruments have improved drastically within the past fifty years, it has come with a cost: the instruments pick up not just the RV data from the planets but from their host stars as well (Rajpaul et al. 2015). Therefore, defining one's own models is necessary, and this can be achieved by fitting various hyperparameters to a given function. The Gaussian Process (GP) is a non-parametric type of regression that offers "best guess" solutions to these hyperparameters.

Rajpaul et al. 2015 describe recent developments in Gaussian fitting for RV data sets, specifically how it can be used in conjunction with stellar activity indicators (H-alpha, calcium IRT, etc.) to better minimize RV perturbations. Using similar techniques, this paper will reveal how a GP was developed to be applied to the DS Tuc Ab RV model.

3.2.3. *Developing a GP Function and Model*

The Gaussian fitting process relies on starting probabilistic equilibrium positions and "moving" in the best direction based on a likelihood function. It bases this

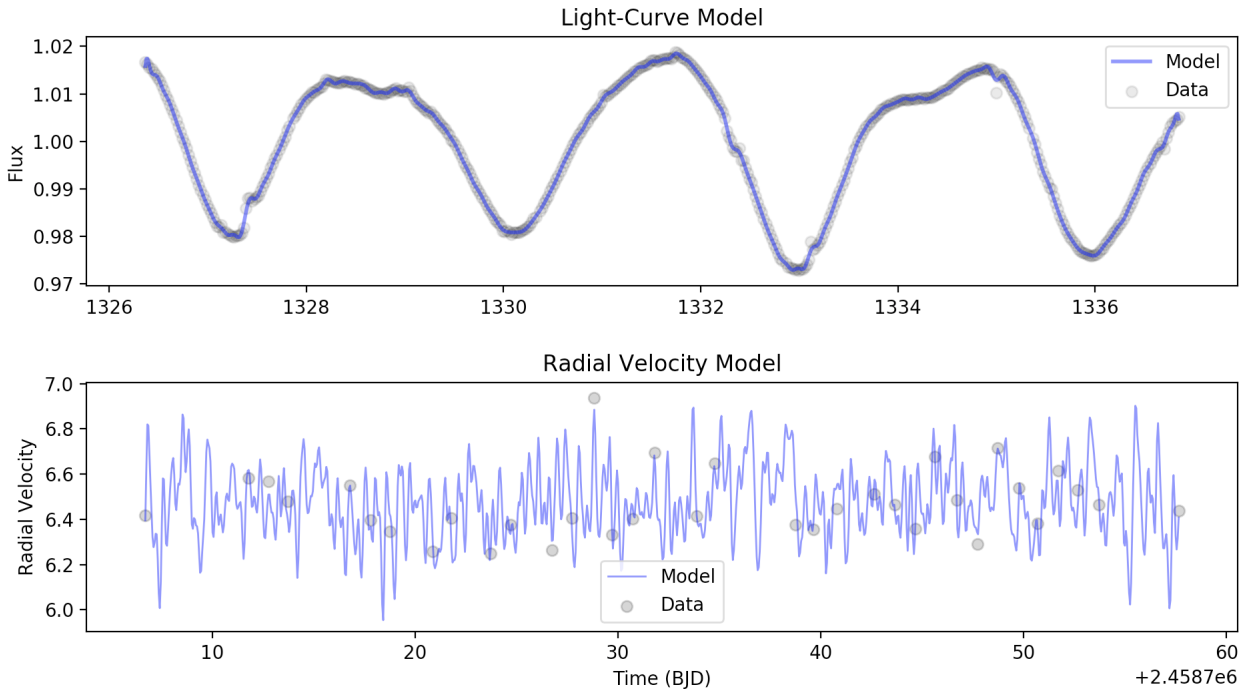


Figure 3. Using the `George` GP regression package, we fit models to the photometry and RV data. The models clearly over-fit the observations, likely due to poorly constrained timescale hyperparameters in the `George` model.

regressive "movement" on a covariance matrix, and the type of covariance is something we can define when writing our GP function.

We can expect to see periodicity in the RV dataset and the stellar activity signals—due to their modulation from DS Tuc A's periodic rotation—so a covariance function with a periodic element (sine) would naturally be fitting as a catalyst for our GP model. An example of a convenient yet powerful covariance, as highlighted by Rajpaul et al. 2015, is the quasi-periodic covariance function:

$$\gamma_{qp}(t_i, t_j) = \exp\left\{-\frac{\sin^2[\pi(t_i - t_j)/P]}{2\lambda_p^2} - \frac{(t_i - t_j)^2}{2\lambda_e^2}\right\}, \tag{4}$$

where (t_i, t_j) is the corresponding RV timescale component of the (i, j) element of the covariance matrix, P is the rotation period of the star, λ_p is the periodic timescale (and is dimensionless), and λ_e is the evolutionary time-scale (has units of time).

With a covariance matrix in place, we can construct the likelihood function, which takes a more complicated form than the one from our linear model:

$$\log(\mathcal{L}) = \mathbf{r}\,\boldsymbol{\gamma_{qp}}^{-1}\,\mathbf{r}^{\mathrm{T}} \;+\; \log\left(\det \boldsymbol{\gamma_{qp}}\right) \;+\; n\log\left(2\pi\right), \tag{5}$$

where $\mathbf{r}$ is the vector of residuals between the RV data and the RV model (i.e. $\mathbf{r} = z - z_{model}$) and n is the number of RV data points.

To implement a MCMC and GP efficiently, and to model our data from the resulting Gaussian posteriors, we utilize a few outside packages, namely `emcee` (Foreman-Mackey et al. 2013), `George` (Ambikasaran et al. 2014), and `RadVel` (Fulton et al. 2018).

3.3. *Fitting with `George`*

`George` is a fast and powerful GP regression library with many different kinds of kernels and fitting models to best match the user's needs. To compute our GP model, we use `George's` non-stationary periodic exponential sine squared kernel, given by

$$k(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\Gamma\ \sin^2\left[\frac{\pi}{P}|x_i - x_j|\right]\right), \tag{6}$$

where $(\mathbf{x}_i, \mathbf{x}_j)$ is an element of our covariance matrix, Γ is a length scale of our correlations (and a scaling hyperparameter), and P is the stellar rotation period.

We use the kernel to compute the likelihood for both the RV data and the photometry data, and then combine those likelihoods to be passed into an MCMC sampler known as `emcee`, which is an effective package for constraining the hyperparameters of our models. Figure 3 reveals the resulting plots from fitting the light-curve (photometry) and RV models to their respective data. We found that the exponential sine squared kernel from `George` over-fits our observations, so we turn to package more specific to modeling RV signals.

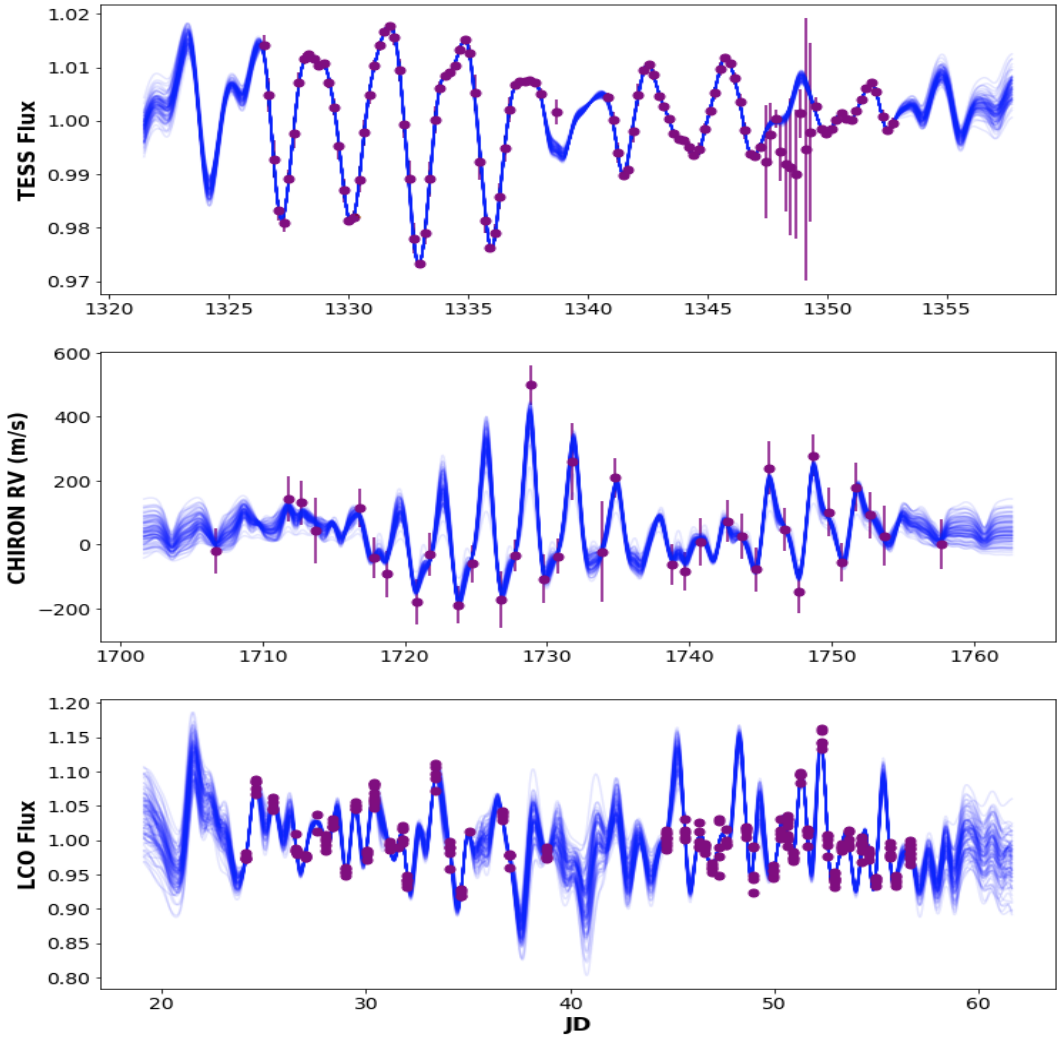


Figure 4. Using `RadVel` and `emcee` to develop a GP fitting process, these plots reveal the resulting models for the photometry data from TESS, RV data, and photometry data from LCO for the DS Tuc system. All three plots demonstrate well-fitted models, which we can now use to do a more in-depth analysis on DS Tuc Ab.

3.4. *Fitting with `RadVel`*

`RadVel` is special in the sense that it is a library specifically built for modeling RV data, which is notoriously noisy and difficult to isolate from stellar activity signals. We therefore take full advantage of its RV-specific capabilities, including built-in RV modeling, covariance kernels, and parameters classes.

We start by constructing a `RadVel` GP model to describe the RV, TESS discovery photometry, and LCO follow-up photometry observations (the calcium IRT data ended up not fixing the RV noise well enough with `RadVel`, so instead the photometry data from LCO was adopted). We then compute the likelihood functions for each model with the radvel.likelihood.GPLikelihood class, passing in the following quasi-periodic function for the covariance kernel (which is very similar to Equation (4)):

$$C_{ij} = \eta_1^2 * \exp\left\{\frac{-|t_i - t_j|^2}{\eta_2^2} - \frac{\sin^2\left(\frac{\pi|t_i - t_j|}{\eta_3}\right)}{2\eta_4^2}\right\} \quad (7)$$

We then combine the likelihoods from the three models into one marginal likelihood and apply a Gaussian prior on it, which takes a chi-squared form nearly identical to Equation (1). Finally, we initialize our hyperparameters with Gaussian distributions and run our MCMC sample using `emcee` to produce posteriors for the parameters we defined earlier (with the radvel.Parameter class).

Figure 4 shows the resulting plots from our `RadVel` implementation. All three models—for TESS flux data,

Table 1. Hyperparameter means from `RadVel` GP model.

Parameter	Value
K_1 (m s^{-1})	7.1 ± 5.5
TESS GP Light-curve (LC) Amp.	0.0079 ± 0.0010
RV GP Amp.	133.2 ± 23.6
LCO GP LC Amp.	0.1 ± 0.020
GP Exp. Length	6.0 ± 0.50
GP Period	3.1 ± 0.021
GP Period Length	0.39 ± 0.031
RV Gamma	40.5 ± 39.8
TESS GP LC Gamma	1.0 ± 0.0027
LCO GP LC Gamma	1.0 ± 0.031
RV Jitter	23.9 ± 18.8
TESS LC Jitter	0.00014 ± 0.00011
LCO LC Jitter	0.011 ± 0.00055

NOTE—These values are the mean values for each parameter (as the GP produces a normal distribution for each versus just a singular value). For use in calculations such as K_1 in the The Radial Velocity Equation (7), we use the entire list of K_1 values from the GP (not its mean value).

RV data, and LCO flux data—are well-constrained, so we will use these as bases for a more detailed analysis of the system, specifically to isolate a mass upper-limit for DS Tuc Ab.

4. RESULTS AND ANALYSIS

Prior to this paper, the mass for DS Tuc Ab was poorly constrained. Newton et al. 2019 predicted a mass of $28^{+35}_{-17}\,M_\oplus$, based on the Chen & Kipping 2016 mass-radius relation. Here, we present an upper mass limit constraint for DS Tuc Ab using the GP model developed from the `RadVel` and `emcee` packages.

We compute the mass of DS Tuc Ab using our RV semi-amplitude K_1 posterior, as per the Radial Velocity Equation, defined by Lovis & Fischer 2010 to be

$$K_1 = \frac{28.4329\ \mathrm{m\ s^{-1}}}{\sqrt{1-e^2}} \frac{m_2 \sin i}{M_{Jup}} \left(\frac{m_1+m_2}{M_\odot}\right)^{-2/3} \left(\frac{P}{1\ \mathrm{yr}}\right)^{-1/3}, \tag{8}$$

where K_1 is a parameter from our GP that has units m/s (as it is the offset of our RV model, see Table 1), e is the eccentricity of the planet's orbit, $\sin i$ is the orbital inclination, m_1 is the mass of the star, m_2 is the mass of the planet, and P is the orbital period.

In our case, we let $e = 0$, $\sin i = 1$, and the quantity $m_1 + m_2 \approx m_1$, since the mass of the star is so much greater than that of the planet, making the planet's mass negligible. We then solve for the upper mass limit of the planet, m_2, using the upper 3σ values (i.e. 99.5% percentile) of the K_1 posterior, and get a result of 83 $M_\oplus$. We plot this value—and the upper mass limit value had a GP not been used—amongst data from over five hundred other exoplanets from the NASA Exoplanet Archive on a radius vs. mass plot, along with radius vs. mass models from Fortney 2007 for 25 $M_\oplus$ core mass planets orbiting at 0.1 AU from their parent stars (similar parameters to those of DS Tuc Ab) (Figure (5)).

We observe that DS Tuc Ab lies in a rather uncharted region for exoplanets in Figure (5), demonstrative of the fact that large Neptune-like planets are rare and scarcely researched. We also note that with the GP, we are able to rule out highly-dense models for the planet, whereas with no GP, we have an upper mass limit upwards of 220 $M_\oplus$.

5. SUMMARY

The RV method of discovering exoplanets has been extremely successful in recent years given the advancement in precision of measuring instruments. What's more, utilizing the RV method can tell us more about the system than simply the fact that it contains a planet. We can learn about the interaction between a planet and its star, and as seen in this paper, uncover the planet's mass and therefore reveal what it might be made of, something not possible with the transit method.

The unfortunate side effect of measuring RV signals with such exactness is that a great deal of noise from the parent star is often picked up, making it challenging to distinguish the true RV data from the planet. This is where sampling RV, photometry, and stellar activity indicators simultaneously becomes extremely valuable, especially in stars as active as DS Tuc A. GP modeling has provided a new avenue in astrophysics to successfully execute this sampling, and we can now develop much more in-depth analyses of systems that were considered hopelessly noisy just a decade ago.

To employ this GP, we first developed Lomb-Scargle periodograms to determine the datasets that best revealed a rotation period of roughly 2.85 days. Then we attempted to fit linear models to these datasets using a simple chi-squared likelihood function, but this failed at accurately representing the RV observations. So we moved onto utilizing modeling packages, the first of which was `George`, where we fit the TESS photometry and RV datasets using a exponential sine squared kernel (and used `emcee` for our MCMC sampling). This was successful for the photometry data, yet over-fit the

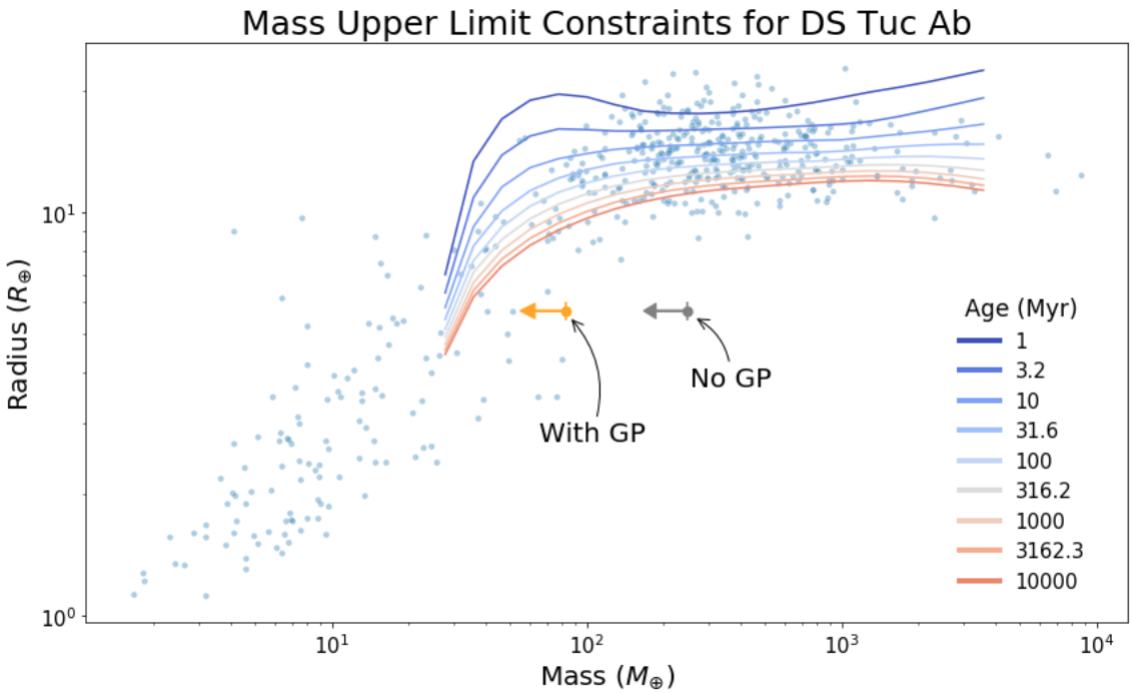


Figure 5. Radius vs. mass plot showing the upper mass limits of DS Tuc Ab with and without a GP constraint (the orange and gray points, respectively). We also plot radius vs. mass data from over five hundred other exoplanets from the NASA Exoplanet Archive (the scatter points), as well as models from Fortney 2007 (the blue-to-red lines), to reveal where DS Tuc Ab lies along the trend.

RV observations. With `RadVel` and a quasi-periodic kernel, however, the fitting process was much smoother, as the hyperparameter posteriors were well-constrained.

With the models in place, we used the Radial Velocity Equation to calculate the upper mass limit of DS Tuc Ab: 83 $M_\oplus$. We then plotted this on a radius vs. mass plot, along with the upper mass limit had a GP not been used, radius vs. mass models from Fortney 2007, and data from roughly 500 other exoplanets.

This is a major step in understanding planetary evolution. Systems as young as DS Tuc are virtually unheard of, and being able to isolate RV signals and a mass limit for the infant DS Tuc Ab planet whilst its radius is still evolving highlights the potential for using GP modeling to learn about stellar activity trends. And with the TESS extended mission underway, the satellite will revisit DS Tuc so we may dive even further into uncovering the mysteries of the planetary past.

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Facilities: TESS, LCO, Exoplanet Archive

Software: `Astropy` (The Astropy Collaboration et al. 2018), `emcee` (Foreman-Mackey et al. 2013), `George` (Ambikasaran et al. 2014), `RadVel` (Fulton et al. 2018).

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